

西方经济学

Part 4 Business Cycle Theory

THE ECONOMY IN THE SHORT RUN

Lecture 4B The Open IS-LM Model

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- (1) M14; S11.3.¹
- (2) 中国人民银行, FED, FED Balance Sheet Trends.
- (3) 其他资料: [GM]
 - THE OPEN ECONOMY REVISITED: THE MUNDELL-FLEMING MODEL AND THE EXCHANGE-RATE REGIME

¹M 指代马工程教材, S 指代课外阅读材料沈坤荣教程。

学习目标

西方经济学

- (1) 掌握开放经济的凯恩斯主义基本分析框架。
- (2) 领会马工程教材精神。



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Outline

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- 1 Case: Small Home vs. Large Foreign
- 2 Case: Large Home vs. Large Foreign
- 3 Appendix: Mankiw
- 4 马工程教材疑难重点



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Case: Small Home vs. Large Foreign I

Assumption 1 (Perfect Capital Mobility [PCM])

Residents of a country have full access to world financial markets.

Assumption 2 (Static Expectations of e)

$$\mathbb{E}_t e_{t+1} = e_t.$$

Assumption 3 (Rigid Prices)

P and P^* are rigid.

Robert A. Mundell (1963) and **J. Marcus Fleming** (1962) assume a small open economy (Home), and assumptions 1–3. Without loss of generality, let $G^I = 0$, $NFP = 0$, $TR = 0$, $INT^G = 0$.



Case: Small Home vs. Large Foreign II

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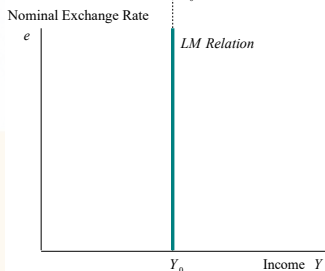
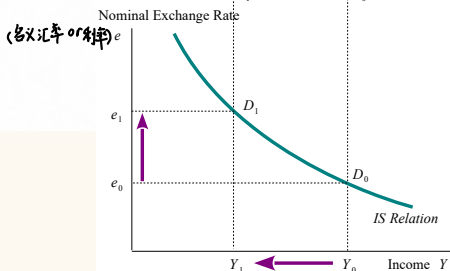
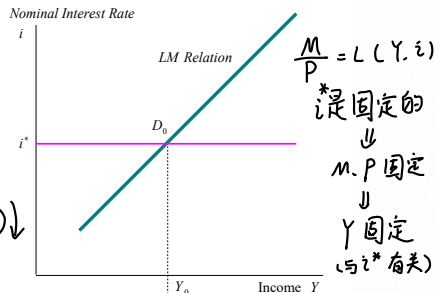
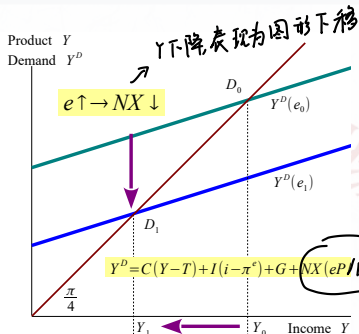
The IS-LM model for an open economy, the Mundell-Fleming model, can be written as

$$\begin{cases} Y &= C(Y - T) + I(i - \pi^e) + G + NX\left(\frac{eP}{P^*}\right), \\ M/P &= L(Y, i), \\ i &= i^*. \end{cases}$$



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Derivation of IS and LM Curves in the MF Model

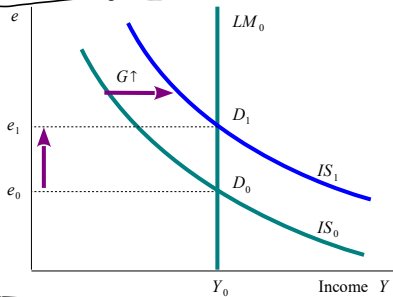


(a) The IS Relation

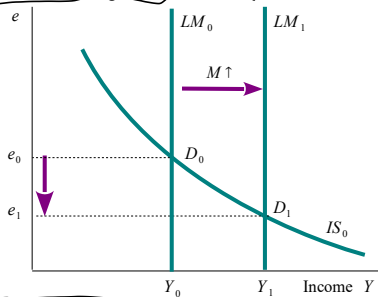
(b) The LM Relation

The MF Model under Flexible Exchange Rates

Nominal Exchange Rate

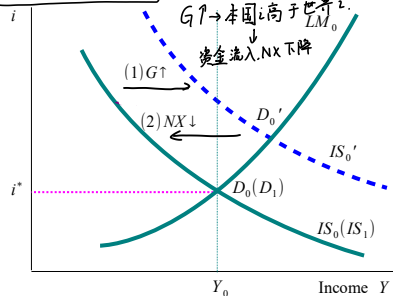


Nominal Exchange Rate

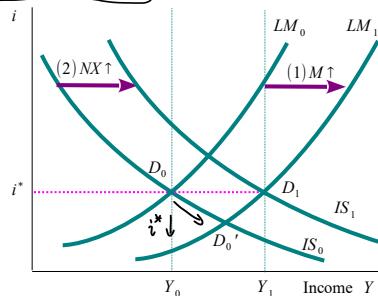


总需求多 IS 向右
总货币供给多 LM 向右

Nominal Interest Rate

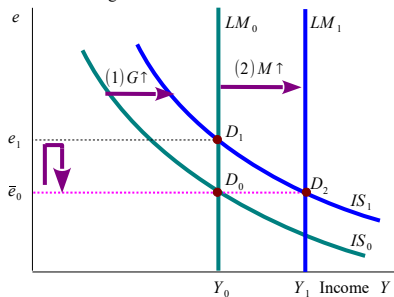


Nominal Interest Rate

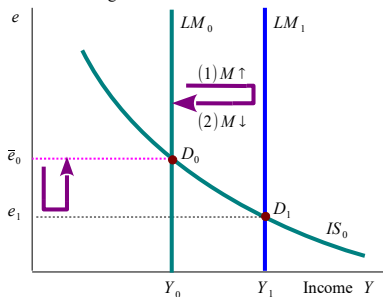


The MF Model under Fixed Exchange Rates

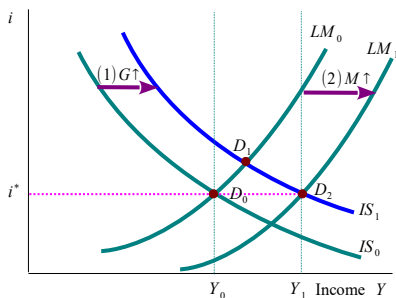
Nominal Exchange Rate



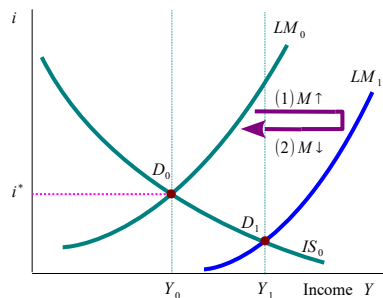
Nominal Exchange Rate



Nominal Interest Rate



Nominal Interest Rate



The Mundell-Fleming Trilemma

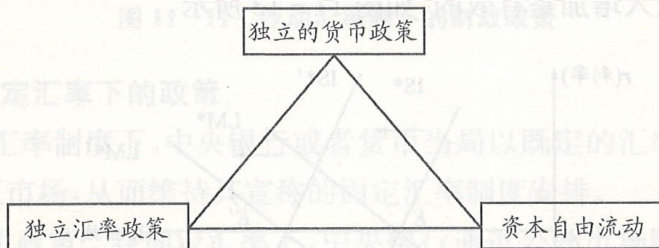


图 11-14 蒙代尔—弗莱明三元悖论

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Case: Large Home vs. Large Foreign

Assumptions 1–3 still hold.

Large Home's IS : $Y = C(Y - T) + I(i - \pi^e) + G + NX\left(\frac{eP}{P^*}\right);$

Large Home's LM : $\frac{M}{P} = L(Y, i);$

Large Foreign's IS^* : $Y^* = C^*(Y^* - T^*) + I^*(i^* - \pi^{e*}) + G^* + NX^*\left(\frac{P^*}{eP}\right);$

Large Foreign's LM^* : $\frac{M^*}{P^*} = L^*(Y^*, i^*);$

Perfect Capital Mobility: $i = i^*;$

The World is closed: $NX\left(\frac{eP}{P^*}\right) + NX^*\left(\frac{P^*}{eP}\right) = 0.$



Case: Large Home vs. Large Foreign

The equations are reduced to the following system of equations.

$$\text{全球 IS} \leftarrow IS + IS^* : \quad Y + Y^* = C + C^* + I(i - \pi^e) + I^*(i - \pi^{e*}) + G + G^*; \quad (1)$$

$$LM : \quad \frac{M}{P} = L(Y, i); \quad (2)$$

$$LM^* : \quad \frac{M^*}{P^*} = L^*(Y^*, i); \quad (3)$$

$$IS : \quad Y = C(Y - T) + I(i - \pi^e) + G + NX\left(\frac{eP}{P^*}\right). \quad (4)$$

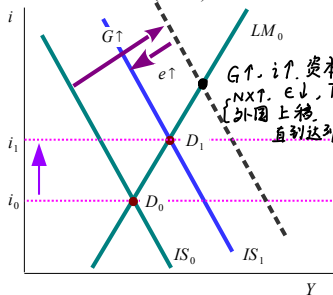
If e is flexible, equations (1)–(4) determine Y, Y^*, i, e .

If e is fixed by Home, equations (1)–(4) determine Y, Y^*, i, M .

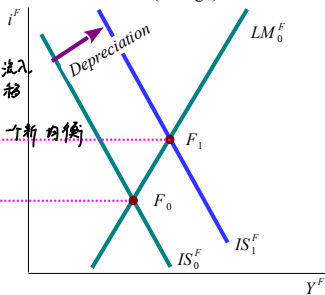


Case: Large Home (Flexible e) vs. Large Foreign (Flexible e^F)

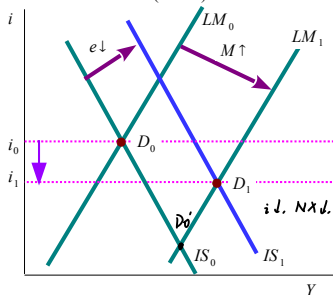
Nominal Interest Rate (Home)



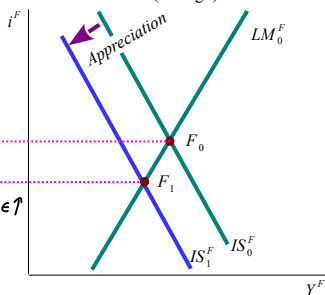
Nominal Interest Rate (Foreign)



Nominal Interest Rate (Home)



Nominal Interest Rate (Foreign)



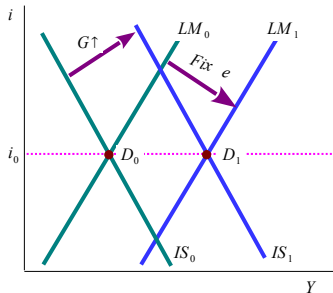
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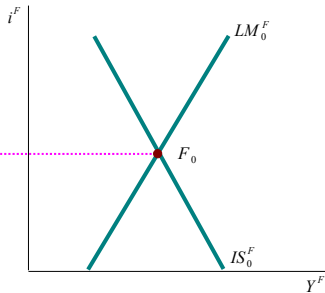
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Case: Large Home (Pegged e) vs. Large Foreign (Flexible e^F)

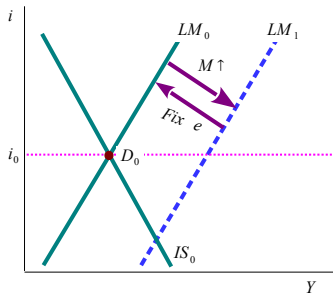
Nominal Interest Rate (Home)



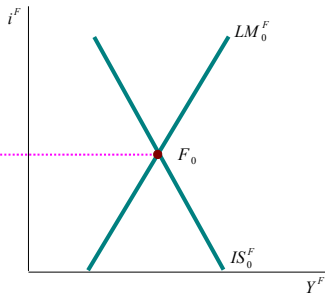
Nominal Interest Rate (Foreign)



Nominal Interest Rate (Home)



Nominal Interest Rate (Foreign)



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Appendix: Mankiw

Small Home–Large Foreign (Imperfect Capital Mobility)

$$\text{IS:} \quad Y = C(Y - T) + I(r) + G + NX(eP/P^*)$$

$$\text{LM:} \quad M/P = L(Y, r + \pi^e)$$

$$\text{ICM:} \quad NX(eP/P^*) = NFI(r - r^*).$$

Under flexible exchange rate regime, the endogenous variables are Y, r, e ;

Under fixed exchange rate regime, the endogenous variables are Y, r, M .

Substituting for NX gives

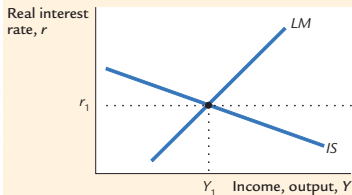
$$\text{IS:} \quad Y = C(Y - T) + I(r) + G + NFI(r - r^*)$$

$$\text{LM:} \quad M/P = L(Y, r + \pi^e).$$

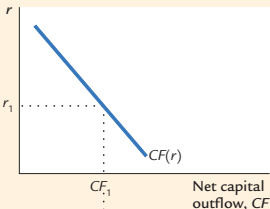


FIGURE 13-15

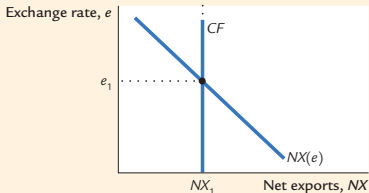
(a) The IS-LM Model



(b) Net Capital Outflow



(c) The Market for Foreign Exchange

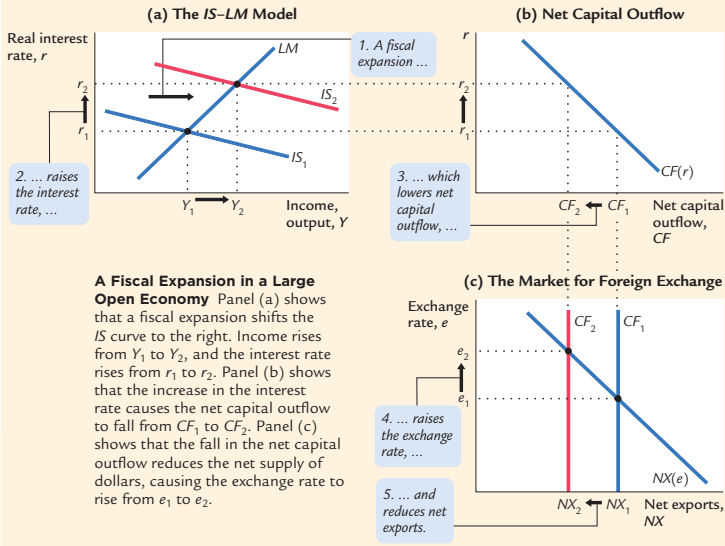


A Short-Run Model of a Large Open Economy Panel (a) shows that the IS and LM curves determine the interest rate r_1 and income Y_1 . Panel (b) shows that r_1 determines the net capital outflow CF_1 . Panel (c) shows that CF_1 and the net-exports schedule determine the exchange rate e_1 .



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FIGURE 13-16

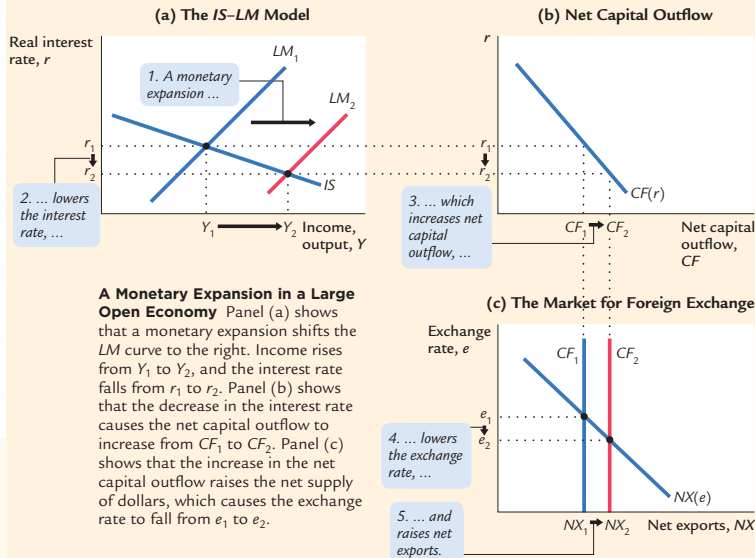


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FIGURE 13-17



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西方经济学

- (1) 在开放经济凯恩斯模型, Y, C, I, S, NX, i, e 如何决定?
- (2) 理解 Mundell-Fleming Trilemma.



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1 (E2, p.202)

根据马工程教材观点，应当如何评析西方经济学的蒙代尔-弗莱明模型？



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